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A Study on the Impact of Screen Curvature and Control Display Gain on Pointing Performance in Large Displays

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BETTER LIFE ERGONOMICS FOR FUTURE HUMANS

01 Introduction

ⓒ LG Display

ⓒ IT Chosun, Samsung 49” ultra wide gaming monitor

- Growth in large display market across various sectors

  ex. In-vehicle entertainment for autonomous vehicles, immersive gaming with curved displays, etc

- Curved displays as a solution to drawbacks of large displays

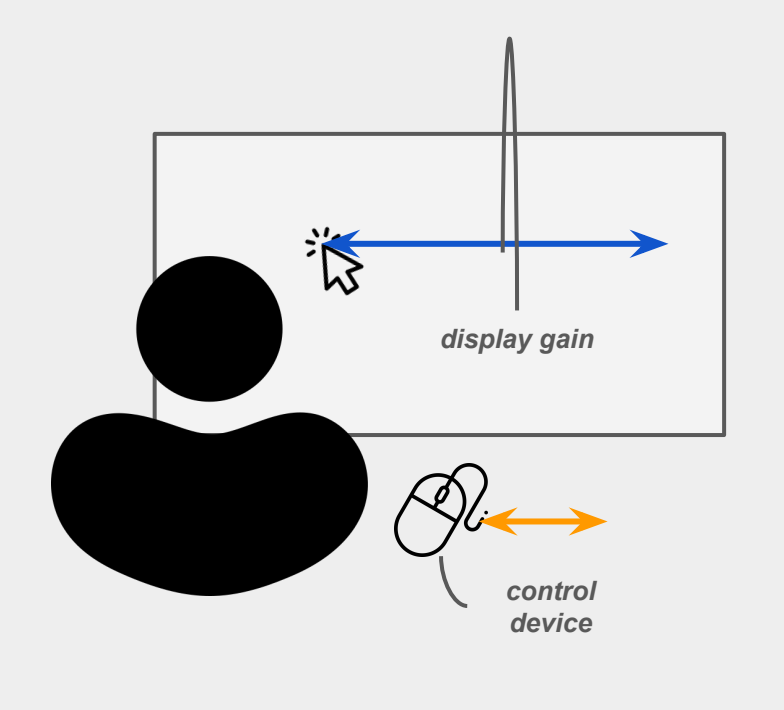
  Providing wider FOV by equal viewing distances across the screen

02 Background

- **Control-Display(C-D) Gain**

- The ratio of the distance moved on the display to the distance moved by the control device.
- Optimal CD gain improves accuracy in the point-select process.

- Due to the speed-accuracy tradeoff, studying optimal control conditions using only the minimum movement time is challenging.

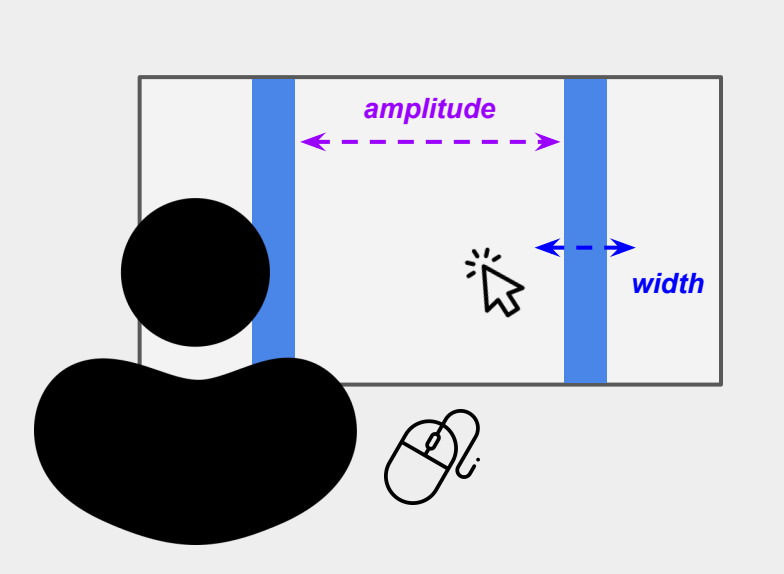


02 Background

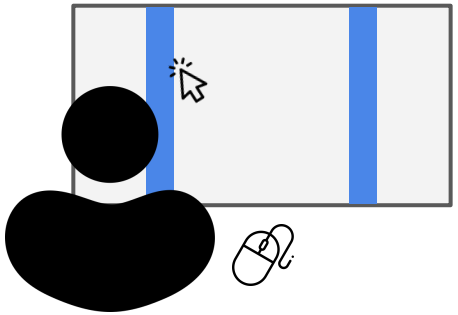
- **Fitts' Law**

- Establish a relationship between **movement time (MT)** and the variables of movement **amplitude (A)** and **target width (W).**

- **Throughput** derived from Fitts' Law includes both speed and accuracy as performance metrics.
- Examine the interaction between screen curvature and mouse speed performance on large displays.



03 Research Objective



- **Identify impact of screen curvature and C-D gain for pointing performance on large displays**

  - Utilizing Fitts' Law to measure user's pointing performance regarding to mouse C-D gain and curvature

04 Methodology

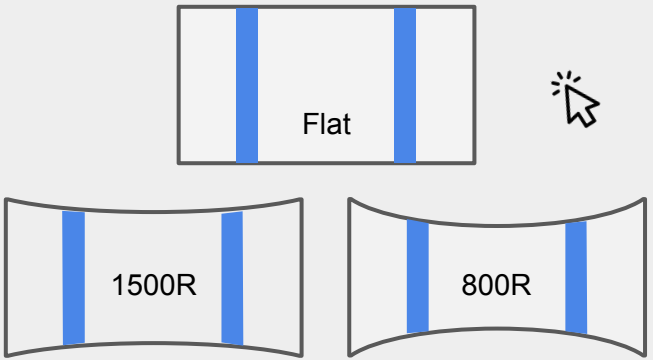
- **Participants**
  - n = 30 (10 females, 20 males)
  - Average 26.2 years old
  - All right handed, competent computer users

- **Experimental Design**
  - 3 * 2 within-subjects
  - IV: Display curvatures (Flat, 1500R, 800R) * C-D gain (small: 4, large: 12.7)
  - DV: Throughput(TP) and Movement Time(MT) derived from Fitts' Law

- **Procedure**
  - FittsStudy program - 12 trials per condition, first 2 trials served as practice
  - Asked to perform Fitts task as quickly and accurately as possible
  - Few verbal questions were asked to collect subjective perceptions

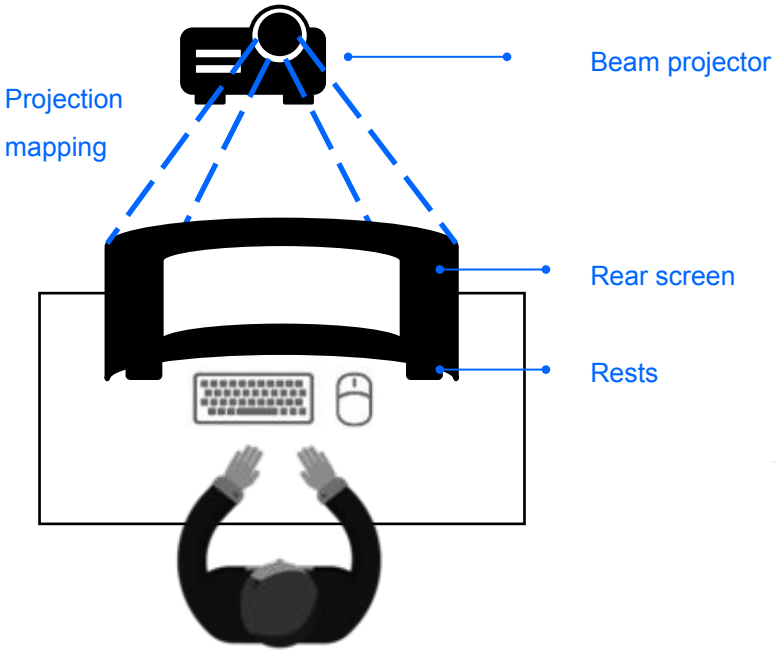
04 Methodology

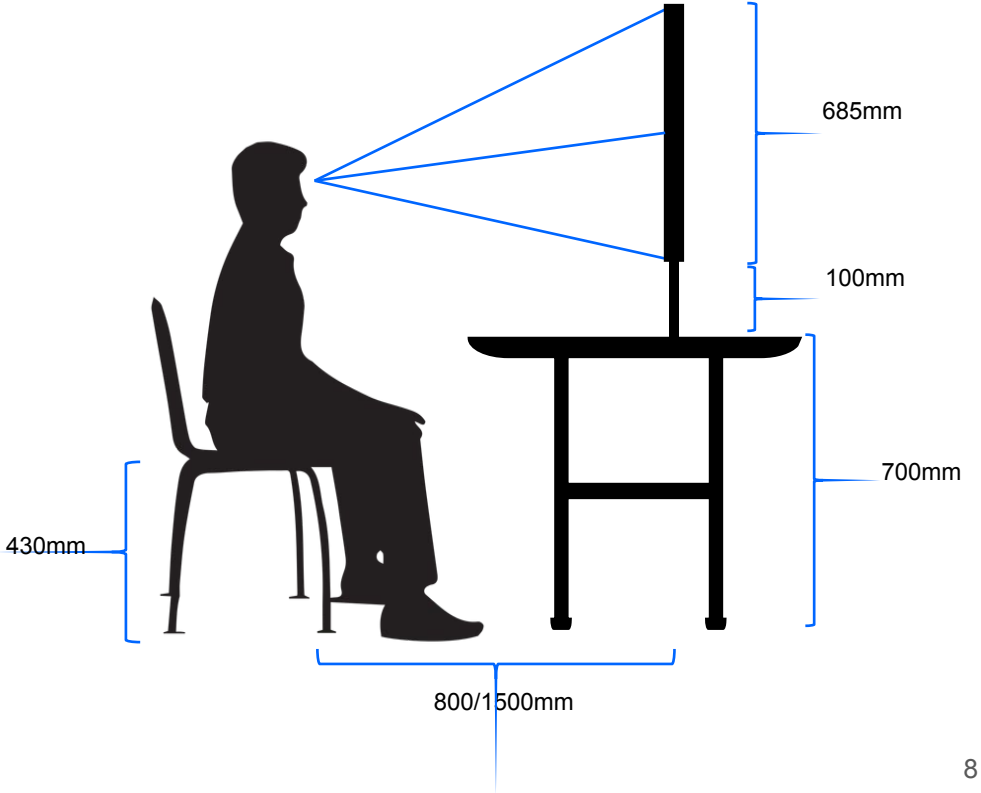
- **Experimental Setup**
  - **Three large displays**
    - Curvature conditions, Flat, 800R, and 1500R.
    - Large displays were 145 cm in length, each with a screen resolution of 1366 × 768 pixels.
  - **Fitts' Law**
    - Demonstrated by FittsStudy Program
    - Target widths: 32, 64, 128(px); Amplitudes: 256, 640, 1024 (px)
  - **C-D gain**
    - Adjusted by configuring the mouse speed settings
    - Small C-D gain: 4 ← Mouse speed: 4
    - Large C-D gain: 12.7 ← Mouse speed: 6



04 Methodology

- **Experimental Setup**





05 Data Analysis

- **Data Acquisition**

  - Target Widths; 32, 64, 128 (px) & Amplitudes; 256, 640, 1024 (px)
  - 8 Index of Difficulty (ID) values were repeated for 12 trials per condition
  - 540 data points per participant * 30 participants = 16,200 data points

- **Data Analysis**

  - Repeated measures ANOVA by SPSS
  - Two calculation approaches were analyzed for Throughput(TP)
    - TP_{avg}: mean-of-means approach
    - TP_{slope}: slope inverse approach

06 Results

- **Movement Time(MT)**

<Summary of MTe>

C-D gain	MeanMTe	SD	F	p-value
Small(4)	1.015	0.005	122.252	<0.001*
Large(12.7)	0.946	0.004		

<Summary of MTe in large C-D gain condition>

C-D large	MeanMTe	SE	F	p-value
Flat	0.964	0.005	7.331	=0.001*
1500R	0.990	0.005		
800R	0.986	0.007		

- Significantly shorter MT with large C-D gain condition compared with small C-D gain condition

  (Mean MTlarge: 0.946, Mean MTsmall:1.015, F=122.252, p<0.001)

- The effect of curvature on MT was not significant
  - However, in large C-D gain, the flat condition resulted in a shorter MT

06 Results

- **Throughput(TP)**

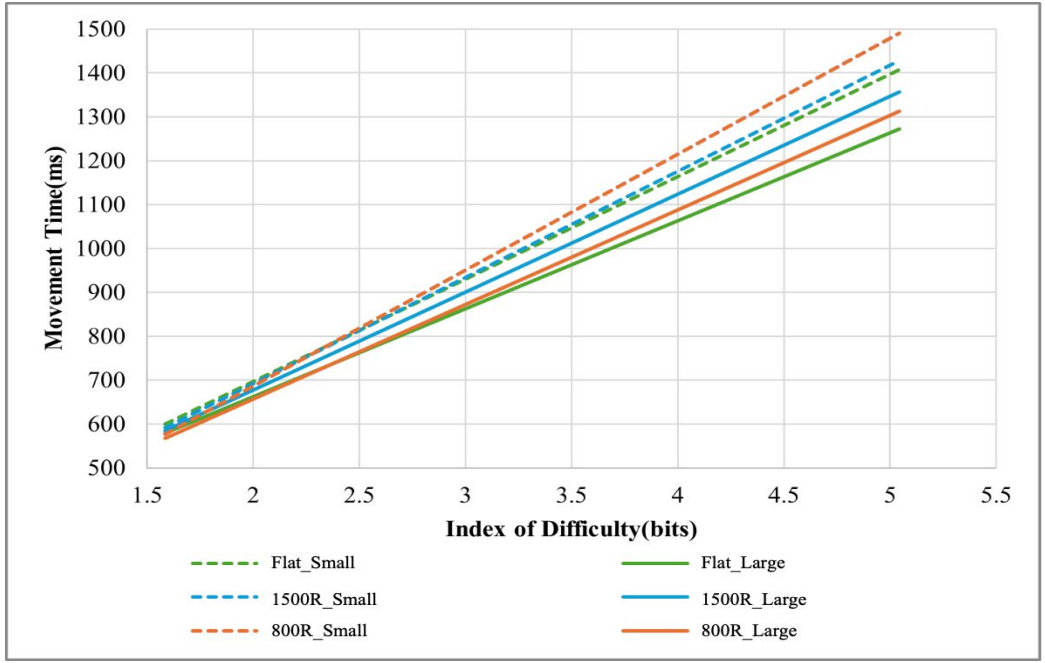
<Summary of TP_{inv} >

C-D Gain	MeanMTe	SD	F	p-value
Small(4)	4.046	.154	28.364	<0.001 *
Large(12.7)	4.959	.188		

- TP_{avg} showed no significant differences between curvature and C-D gain.

06 Results

- **Modeling Movement Time with ID**

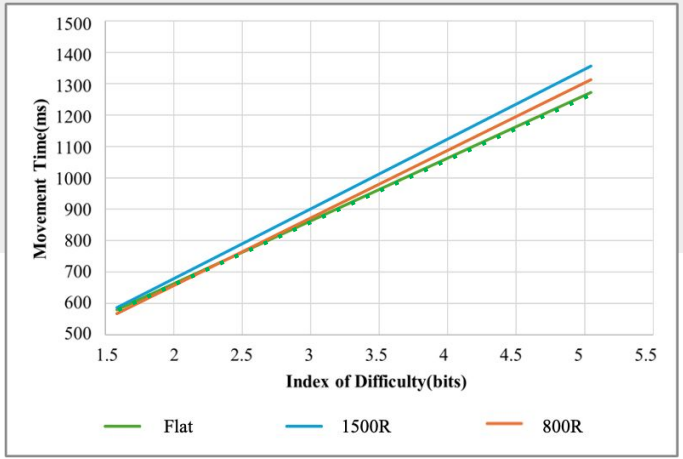


- As C-D gain increases, the impact of C-D gain becomes more prominent, with smaller C-D gain condition resulting in longer movement time(MT)

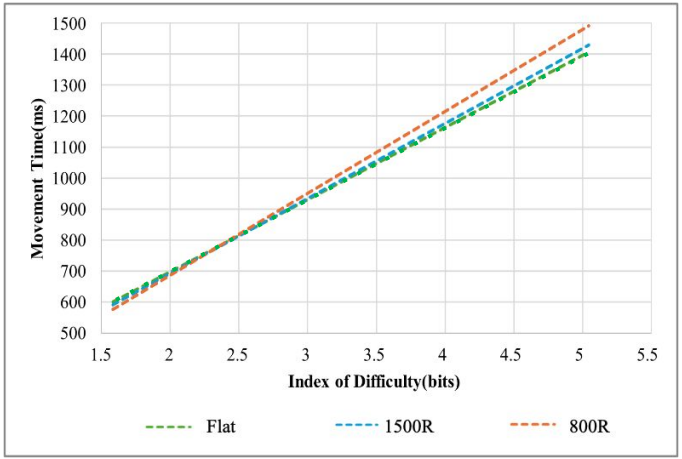
06 Results

- **Modeling Movement Time with ID**

- A noticeable trend indicating that the flat display condition showed lower MT compared to curved display conditions.
- ***"It(Flat) is more comfortable as I'm more accustomed to it"***



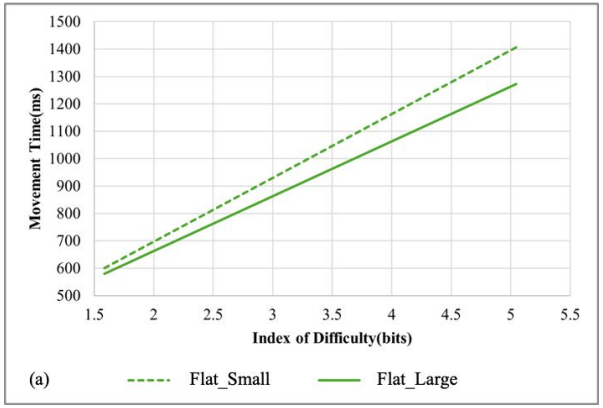
<Large C-D gain condition>

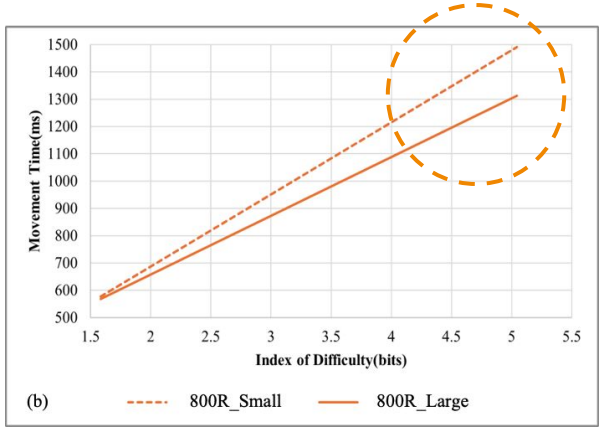


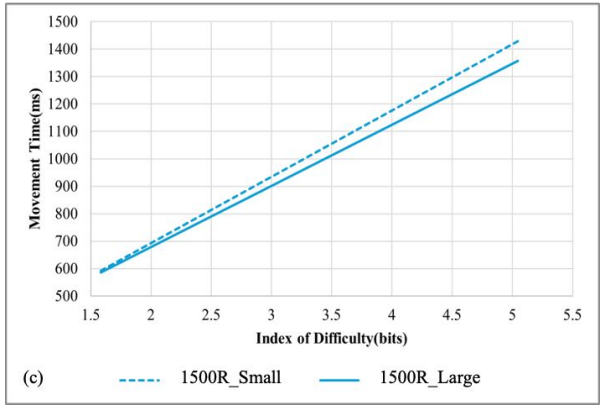
<Small C-D gain condition>

06 Results

- **Modeling Movement Time with ID**





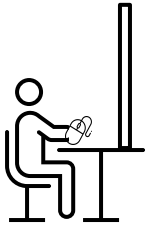


- The largest discrepancy observed at higher ID levels under 800R condition
- ***"When there is curvature, the speed seemed to move faster than I intended at the curved parts, so it was most difficult to control when the curvature was severe."***

06 Results

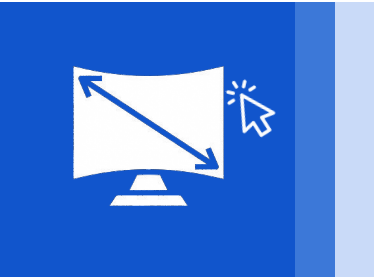
- **Subjective Evaluation**

- Some participants found it easier to control the mouse at higher speed, while others found it inconvenient to move the mouse multiple times at slower speed.



- Several participants noted that turning their necks was easier with curved displays, while they found it visually inconvenient with the flat display when the edges of the screen were not visible at a glance.

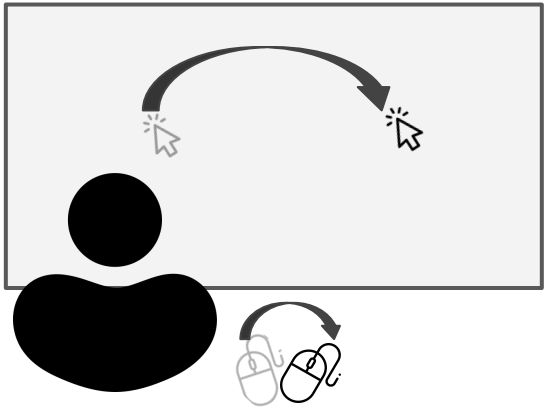
07 Discussion



- **Main effect of C-D gain in pointing performance**
  - Large C-D gain rather than small might be more effective in large displays considering pointing performance

- **Consideration of curvature in large displays**
  - Difference in pointing performance across curvatures
    - As screen curvature increases, a distortion effect may lead participants to underestimate the distance they have traveled.
  - Curved displays might have more impact on subjective user experience, while throughput was higher with flat display.

08 Conclusion



- **Effects of curvature and C-D gain on the performance of Fitts task in large displays.**

  - Significant effects of C-D gain and the presence of curvature in large C-D gain condition on performance were revealed.

  - The result suggests that higher C-D gain may be preferable for large displays, and curvature influences control performance.

  - The present study is expected to be applied and utilized in various future studies addressing large displays.

Thank you

QnA time!